Validating Eyes-free Affordance of On-Finger Hand Proximate User Interfaces in In-situ Scenarios

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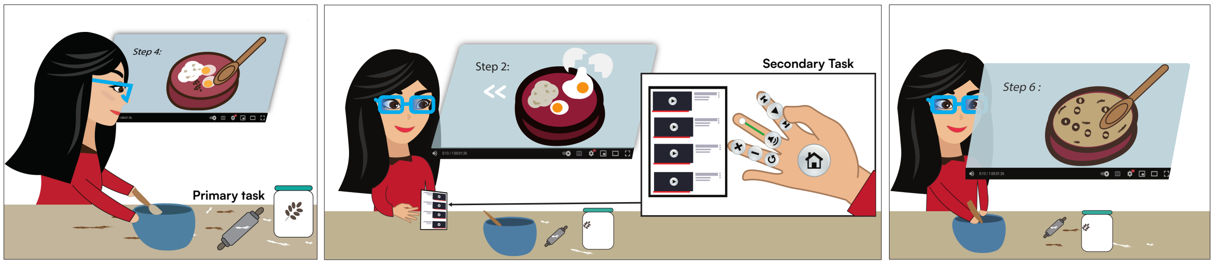

Figure 1: An example in-situ activity, where a Hand Proximate User Interface (HPUI) can support the user's primary task (cooking) by providing eyes-free support. In this scenario, Cloe is following a cooking tutorial on how to make a dish while wearing an HMD. She follows the video and also interacts with the HPUI, which is a secondary interaction or task, to play, pause, and rewind the video then go back to cooking. When she further familiarize herself, she can quickly interact with the HPUI and make accurate selections without having to look at her hand. As a result, she can spend more time observing her core activity.

Abstract

We explore the value of Hand Proximate User Interfaces (HPUIs) for in-situ interactive head-mounted systems, i.e. systems designed to support a user's primary activity. HPUIs are unencumbered, single-handed, and have tactile and proprioceptive affordances. This allows novice and expert users to rely on the visual cues available in the UI, but to then gradually interact eyes-free such that they minimize interrupting the user's core task. Prior work on HPUI falls short of validating this premise. We address this gap with comparative analysis in the context of a compound task, where one hand is involved in a primary task and the second interacts with a secondary but supporting task (e.g. selecting items from a menu). We compare HPUI with mid-air direct interactions, a common form of interacting with head-mounted displays. The results show that HPUI performs similarly to mid-air but with better eyes-free affordances and user preferences.

Index Terms: in-situ interactions, mid-air interactions, hand interactions

1 Introduction

HMDs have long-promised applications that seamlessly integrate with day-to-day tasks. They boast an infinite canvas to display content and numerous input modalities are available at their disposal. We ask the question of how well can existing HMD interaction techniques support a user's need in their primary activity. For instance, in in-situ applications like Cloe's example in Figure 1, users often need to navigate menus and UI widgets while engaged in a task. Such applications have been widely explored in the literature, such as for interactions while driving semi-autonomous vehicles [43], in medical applications [50] and, for immersive data exploration [1, 16]. Such interactions are common in virtual reality as well - where one interacts with the world around them while simultaneously interacting with menus. Any interaction technique that supports such in-situ use cases must not interfere with the user's primary activity, while also being comfortable and efficient [22, 3, 15]. The properties of Hand Proximate User Interface (HPUI) could make it an ideal candidate for interaction in such application scenarios.

HPUI is a family of interfaces for Head Mounted Displays (HMDs) that display digital content on and around the hand, and interact with them using thumb-to-finger interactions [18, 19]. They are unencumbered and comfortable, both physically and socially [19, 40]. The visual feedback lowers the barrier for novice users - akin to smartphones, they merely have to look at the hand and tap. When considering the on-finger HPUI elements as Faleel et al. [18] had defined, advanced users can take advantage of the proprioceptive and tactile affordance, which would allow them to build memory and interact with the menus without the need to look at their hands. For in-situ applications, such as Cloe's example above, this could prove to be valuable as one could instinctively interact with such menus without diverting their gaze from the primary task. While, prior work on HPUI has used such examples to motivate its value [21, 40, 58] they fall short of validating this claim. Our work is aimed at addressing this gap. In particular, we assess the following claim: in the context of in-situ applications, HPUIs would require less attention to interact with 2D user interfaces such as menus, allowing the user to instead dedicate their attention to the primary task. Here we focus on the on-finger interaction and not the off-finger interactions as described by Faleel et al. [18], as we are interested in the tactile affordance of HPUI.

To address this question, we conduct a within-subject user study with an abstract in-situ compound task - where the primary task inherently requires attention and the secondary task involves inter-

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acting with a 2D menu to progress in the primary task. We expect the task to be similar to what Cloe would do when following a cooking tutorial as described in the above example (see Figure 1). We compare HPUI to mid-air direct interaction, a *defacto* technique for interacting with HMDs, where selection is done by "poking" at targets with fingers. While many factors influence the applicability of HPUI for in-situ interactions, we focus on the following: (1) Users would be more efficient, (2) more comfortable, (3) pay less visual attention to the menu and instead focus on the task at hand, (4) which hand is preferred to interact with the menu. Results from our user study (N=16) show that HPUI is at least as efficient as mid-air direct interaction, and is significantly better with common interactions (i.e., interactions that are frequent and easy to remember/learn). HPUI is also shown to require much lesser attention, i.e., once familiar, participants look at the hand less and pay more attention to the in-situ activity. Further, we also explore handedness as a factor - i.e., the hand that is used to interact with the menu. The Dominant Hand (DH) is better in many contexts [2, 13], yet, prior research has used the Non-Dominant Hand (NDH) for micro gestures with bi-manual tasks [11, 60]. Also, the performance and preference of handedness have received less attention for HPUI, which motivated us to evaluate the handedness (DH vs NDH) as well. Results also show that the DH is preferred when interacting with menus. Our main contribution is a within-subject comparative analysis that provides empirical support for the utility of HPUIs with in-situ tasks and the handedness for interacting with menus.

2 Related work

In this section, we briefly discuss HPUI and its characteristics and also the interaction techniques for in-situ interactions. In particular, we focus on unencumbered interactions, though we expand on other input modalities where applicable.

2.1 Interaction techniques for in-situ interactions with HMDs

In this section, we briefly discuss the different input modalities for 2D content available for HMDs in the context of in-situ interactions. As emphasized by prior research, any interaction technique that is to be used in-situ must be unobtrusive, non distracting, comfortable, and efficient to use [36, 22, 17, 16, 48]. But when one considers the input modalities available with HMDs they have trade-offs between these factors [36, 32]. Consider the use of gestures on HMDs - with proprioception, interactions can be performed without requiring visual attention. Yet, they may not be particularly comfortable, physically [25] and socially [6]. Microgestures, on the other hand, can be more comfortable [6, 19]. However, the lack of visual feedback makes gestures hard to discover and adopt by users [5]. Mid-air interactions, where users would directly interact with elements with their hands, have similar characteristics to HPUI, albeit without tactile feedback. Particularly, when positioned relative to the body [26, 38, 56, 57]. It can utilize proprioception to select targets [38, 56], and can support eyes-free interactions as well [57, 26]. However, the lack of tactile feedback makes it hard to know when an interaction has happened [38, 54]. Also, positioning content for mid-air interactions is not a trivial problem when considering factors like comfort [3, 25] and spatial transitions [37]. Whereas with HPUI, content is always anchored to the hand.

Mid-air pointing, is closely related to mid-air direct interactions. As it is commonly used to interact with elements out of reach, 2D menus would commonly be placed at a distance, with a raycast from the hand moving a cursor, and selection done with a gesture such as pinch. Cockburn et al. [9] show that learnability as a result of proprioception, would be similar to mid-air direct interaction but has a lower accuracy.

Interaction techniques such as gaze and voice [42, 35], while well-suited for some applications, have some drawbacks when used in-situ. Gaze interaction can lead to the Midas touch problem [29], which requires a different modality to augment it, such as pinch [41, 51]. Another limitation of gaze-based input is it inherently requires visual attention, which can be a drawback for in-situ tasks that favor eyes-free interactions. With voice, while it allows expressing complex interactions [42, 35], it is socially uncomfortable [14] and would not be ideal for rapid repeated interactions [35].

HPUI and its characteristics are further expanded upon in Section 2.2. We consider mid-air direct input as the closest to HPUI in terms of affordances, hence, we compare HPUI to mid-air direct input in our user study.

2.2 Hand proximate user interfaces

The design space of HPUI [18] is based on thumb-to-finger interactions [47, 28, 49], which are a subset of microgestures [6]. It is unimanual and unencumbered. Similar to microgestures, they do not suffer from the limitations of gestures such as fatigue [3, 25] or Midas touch problem [55]. Further, it is more private and comfortable to perform, both socially and physically compared to some of the microgestures [19, 40]. Faleel et al. [18], had considered both targets directly on the finger surface and also off of the finger, where the thumb is poking a virtual target but not making contact with any of the fingers. For example, it could be a target between two fingers and the user has to poke the thumb between the fingers to select it. However, here we are only considering the on-finger interaction, as it affords better tactile interactions.

The concept of HPUI itself had been proposed in previous research; for example, Bowman et al. [4] explored the idea of displaying menus on each finger in virtual reality and interacting using the pinch glove. But they were coarser interactions, due to the lower resolution of detecting thumb-to-finger interactions. It is worth noting that Bowman et al. [4], while expecting more eyes-free interaction, did not observe any, potentially as the experimental setup did not focus on it. Recent work on improving the detection of thumb-to-finger interactions [31, 34] has made it possible to have subtler interactions even with currently available consumer devices [18]. With vision-based hand pose estimation improving [46], it would be possible to integrate HPUIs with existing HMDs without augmenting the hands.

An advantage of HPUI over microgestures is its functions do not have to be imagined or remembered. Any interactive element would be displayed on the hand and can be interacted with the same way one would with smartphones. But unlike smartphones, with HPUI, proprioception, and tactile cues would allow users to gradually build memory [24], similar to the use of physical keyboards [59, 45]. This could allow HPUIs to be preferable in in-situ tasks. However, this advantage has not been empirically validated. On the other hand, Perella-Holfeld et al. [40] also identify other potential limitations. Namely, the interactive surface of HPUI is limited and is also a non-contiguous dynamic surface, which can negatively impact user experience. This is in contrast to how interfaces would appear with mid-air direct input. The results from our study aim to shed light on this trade-off.

2.3 Handedness for secondary interaction

Most in-situ applications proposed for HPUI in prior work can be considered to be asymmetric bimanual tasks [18, 21, 58]; where each hand does something different, with at least one hand being used to display and interact with content. Guiard [23] proposed a model for bimanual interactions following the observation that many real-world tasks involve two hands doing different things. Consequent research on Human-Computer Interactions has demonstrated the advantage of designing interfaces that consider this [30, 39]. Prior work also emphasizes that not all bimanual tasks are optimal [30], when not designed well, they could be similar to "tapping the head while rubbing the stomach". Wambecke et al. [52] demonstrate the use of microgestures for a compound task in a flight cockpit. However, we cannot generalize the result to HPUI because of its visual component. The unique visual design of HPUI [40] can be less intuitive, or conversely, it can have a better user experience as it is a direct input modality [33]. To the best of our knowledge, prior work on HPUI has not validated its applicability in the context of a task.

Another related component to such bimanual interaction, that is not well explored in literature is which hand should be used for interacting with the world and which with the menus. Research on handedness has shown that for finer tasks, the dominant hand generally is more preferred and performant [2, 23]. However, related work that uses microgesture interactions using a glove and a pen for precise interactions - a task similar to ours (see Section 3.2) - uses the dominant hand for the precise interactions, and the non-dominant hand for the microgestures [11, 60]. However, gesture interactions used by them were coarser interactions, such as a pinch with each finger having only one function [11]. Given the more granular interaction on HPUI, it is unclear if HPUI would also be more performant/preferred when on the non-dominant hand in such bi-manual tasks.

3 In-situ Comparative Study

This study evaluates the eyes-free affordance of HPUI in an in-situ task. As a baseline, we are comparing on-finger HPUI to mid-air direct input. As outlined in Section 2.1, HPUI and mid-air share many common elements, such as visual cues, and being able to rely on proprioceptive cues. We also test which hand is preferable to interact with a menu. We use a compound task in our study [30]. This different from a dual task [53, 27], which involves two tasks that are independent. Whereas a compound task is composed of tasks that can be dependent. Note that, in our study, we are referring to visual attention as we are focusing on the eyes-free affordance. We use an abstract compound task that requires one hand to interact with the environment (primary task), and the other to interact with a menu (secondary task) while requiring visual attention on the primary task. We explore the following factors: efficiency, comfort, attention requirement and which hand to use for HPUI while interacting in an in-situ task in comparison to mid-air interactions.

More precisely, the research questions are as follows - In the context of in-situ tasks where one has to pay attention to a primary task, and a secondary task of interacting with a menu,

RQ1 Is HPUI more efficient than mid-air direct input?

RQ2 Is HPUI more comfortable and preferable than mid-air direct input?

RQ3 Does HPUI require less visual attention than mid-air direct input?

RQ4 Is the dominant hand more efficient and preferable for the secondary task?

3.1 Study design

For each technique, handedness is treated as a condition. That is the hand with which the participant interacts with the menu. We treat the handedness as dominant hand (DH) or non-dominant hand (NDH), which is determined based on the demographics questionnaire. Hence we have 4 conditions (2 *techniques* (HPUI, mid-air) x 2 *handedness* (DH, NDH)). The study is counter-balanced across the 4 conditions using a Latin square. For each condition, we have seven blocks. Here we treat the first five blocks as training and the last two blocks are used for analysis - however, there is no change in the task between the seven blocks within a condition. The participants complete all blocks of one condition before they proceed to the next. In each block, there are seven sets, where one set is comprised of four trials, as described in Section 3.2.

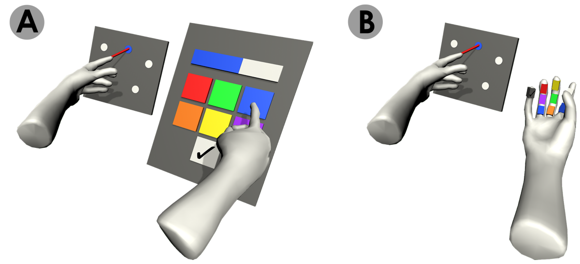


Figure 2: Representation of the task described in Section 3.2. Shows the right-handed condition for both mid-air (A) and HPUI (B) conditions.

3.2 Task

Here we first describe the task the participants performed followed by the rationale. It is a compound task - for a given condition, they have to put a peg in a hole with one hand and interact with a menu with the other hand. Putting the peg in the hole is treated as the primary task whilst interacting with the menu is treated as the secondary task. Both primary and secondary tasks are coupled, i.e., to complete the primary task the participant needs to interact with the menu.

The menu contains a slider, six color buttons, and one accept button (see Figure 2). The holes (targets) are in a workspace separate from the menu. The trials are grouped into sets of four, which we refer to as a set. The holes of the corresponding trials will be randomly placed within the workspace. The target of the active trial would be in one of the colors on the menu's color buttons, while the rest are a neutral color. The peg the user should put in the hole would attach itself to the tip of the index finger of the hand with which the participant would be placing the peg in the hole. At the beginning of each trial, the color of the peg will be set to a neutral color. To complete a trial, the participant has to:

1. Ensure the zoom level of the workspace is within the window where the holes are open. This is done by adjusting the zoom slider to be within this window. That is, if the level is above this window or below this window, the hole will be closed. The location of this window is randomized for each set. At the beginning of each set, the location of the zoom slider is randomly set to a level outside this window.
2. Ensure the color of the peg is the same as the active target. Selecting any of the color buttons on the menu will change the color of the peg.
3. Select the accept button while the peg is placed in the hole.

Once a trial is complete the corresponding target will disappear. When selecting the accept button, audio feedback is also provided to indicate if the trial was successful or not. When all trials of a set are completed, the targets of the trials in the next set are displayed. Note that, for a given set, when the zoom level is within the correct window, the participant does not have to adjust the zoom level for the remainder of the set. But, if they misselect the slider and the zoom level changes, they will have to reposition the slider so that the holes are open.

The intention with this task design is so that the primary task performance can be considered independently,i.e., it would be similar if they were only paying attention to the peg and the hole. However, the secondary task here, by design, interferes with the primary task as it could require visual attention, which in turn slows down the primary task. Hence, what we measure in terms of performance difference would be the effect of this interference.

We include the slider and buttons to observe discrete as well as continuous interactions. The function of the slider and accept button would be the same throughout the entire study. They are also inputs that the participant can easily understand and remember. Whereas the color buttons would require more attention and cognitive load. This distinction can be seen as analogous to interacting with common widgets like the home button/gesture on a smartphone compared to selecting an item on an uncommon interface (e.g., clicking on a link on a website). Note that for all three functions, *Zoom*, *Color* and *Accept*, the participant has to pay attention to the workspace to ensure they are making the correct selection. As the participant would have to further pay visual attention to ensure the peg is in the hole, we expect the attention requirement of the different types of inputs to be further pronounced in the results. That is, when the participant is not familiar with the input mappings, they will have to switch visual attention between the menu and the workspace more often. As they get more familiar, they spend less time directing their attention to the menu.

Before each block, we allow and encourage the participants to reposition the workspace. As we do not enforce a position for the HPUI, we expect the participant to assume a posture that is both comfortable and convenient. A similar treatment is given to the mid-air menu, where they are allowed to move the menu to a desirable position before the blocks, similar to the workspace. Also, note that the participant is seated in a fixed position. Hence, the mid-air menu will be positioned relative to the body of the participant.

3.3 Apparatus and implementation

Figure 3: Setup of the task, where the participant wears a marker glove with the Oculus Quest Pro headset and is seated in a Vicon motion capture volume.

The setup by Faleel et al. [18] is adopted for this study where the hand is tracked using the Vicon motion capture system. Each hand has a marker glove as seen in Figure 3. In this study, we use the Oculus Quest Pro [1] as it allows tracking the gaze, and the study application is implemented with Unity. It is executed using Oculus Link to avoid any performance penalties.

The initial position of the mid-air menu is in front of the corresponding hand. The mid-air menu is tilted by 10 degrees. The workspace is positioned roughly in front of the shoulder. If they are to use the right hand for interacting with the workspace, it would be in front of the right shoulder and vice versa. These positions are determined based on parameters estimated at the beginning of the study during the calibration phase where they are asked to sit upright, with the hands in an L shape, and shoulder width apart.

The slider uses an absolute mapping, i.e., the maximum level of the slider corresponds to the maximum zoom level, and vice versa for the minimum value. Compared to relative mappings such as clutching or joystick-like interactions, this would allow participants to adjust the widget to the desired level quickly.

[1] https://www.meta.com/quest/quest-pro/

At the beginning of a condition, the location of the colors is randomized and would remain fixed throughout the blocks of a given condition. We also ensure the same colors are not seen in two consecutive conditions to avoid any learning effects (see Appendix A in supplementary material [2] for more details).

Figure 2 shows the HPUI menu as well as the mid-air menu. For the mid-air menu, the zoom increase is always from left to right, which is similar to most slider widgets in applications, on HMDs as well as other devices. For the HPUI, the zoom increase is from index proximal phalanx to index distal phalanx, for both hands. Further, the accept button, on the HPUI, is placed on the distal phalanx of the pinky finger as it is more comfortable [18, 12, 28]. The slider on the HPUI is implemented as a deformable interface similar to Faleel et al. [18]. However, here we use Unity's SkinnedMeshRenderer [3] as opposed to the spline-based implementation used there.

3.4 Procedure

After signing a consent form, participants were asked to be seated in the motion capture volume with marker gloves. The Vicon tracking is first calibrated for both hands. Then they are asked to complete a demographic questionnaire while the calibration process is taking place to reduce the overall time of the study. We explain the study task and procedure to them. The participants are asked to be as fast and accurate as possible. Once done, they are asked to wear the Oculus Quest Pro headset, and the study application is launched. They are asked to take the posture to determine the initial parameters for the initial position of the mid-air menu and workspace. Once this is recorded, they are then given a practice session so that they understand the interaction techniques and the task. Then the participants are instructed to perform the task. They are also allowed and encouraged to reposition the workspace and the menu during the mid-air conditions before each block. Between each block, the participants are encouraged to take a rest. After completion of each condition, they are asked to take the headset off and answer a questionnaire. The questionnaire is based on the NASA-TLX (see Appendix B in supplementary material[2]). Once completed they are given as much time as they need to feel relaxed before starting the next block. Once all conditions are completed they are given an exit questionnaire.

3.5 Participants

Sixteen participants were recruited (2 Females, age range 21 - 32; $\bar{x} = 26.7, sd = 4.16$). Two participants (13.33%) reported that their dominant hand was left, and both participants indicated they use their left hand when interacting with the smartphone. Further, 14 participants (93.33%) reported they have some experience with using HMDs as well as hand tracking.

3.6 Measures

We tracked the following measures during the study. Here also we emphasize that we are focused on visual attention as we are interested in eyes-free affordance:

Completion time The time to complete a set, trial, and also the individual functions (Zoom, Color, and Accept)

Attention-percentage The total duration of participants' gaze on the workspace during a trial/function, normalized by dividing it by the total time to complete a trial/function.

Attention switches Number of times the participants' gaze shifted away from the workspace and vice versa, i.e., number of times visual attention switched to and from the workspace.

Errors The number of errors made during a trial.

[2] Suplimentary material also available at https://osf.io/6pej4

[3] https://docs.unity3d.com/Manual/class-SkinnedMeshRenderer.html

Subjective rating With a questionnaire based on the NASA-TLX (see Appendix B(in supplementary material[2])

3.7 Results

Data for each measurement was explored for their normality first. Note, throughout the analysis, appropriate types of analyses were selected after checking the assumptions (i.e., Parametric vs. Non-parametric). Table 1 shows the summary of the results. For brevity, in this section, we only refer to relevant results. The analysis is conducted on the last two blocks. For completion, we also analyzed the first and last blocks, but the results were as expected, where the last block was significantly better than the first block, hence results are included in Appendix C (in supplementary material[2]).

3.7.1 Completion time

We first examined the completion time of a single set. Figure 4 shows the results as the blocks progress. After removing outliers using 1.5 IQR (or Inter-Quartile Range), we first conducted two-way repeated measures ANOVA on completion time to evaluate the effect of technique and handedness. Table 1 shows the results under *combined* completion time. Handedness has a significant main effect on the completion time, where the DH ($\bar{x} = 10.61s$) was quicker than the NDH ($\bar{x} = 11.18s$). Though no significant effect or interaction was found with technique, HPUI DH has an overall lower completion time.

We then individually analyze the time spent on the three functions *Zoom*, *Color*, and *Accept*. Note that the Color and Accept functions were analyzed from all trials. However, the zoom function is expected only during the first trial of a set, hence analysis of Zoom is only with the first trial of each set. We conducted two-way repeated measures ANOVAs (techniques $\times$ handedness) for all three functions on the average time spent on each function by each participant. Table 1 shows the results for three functions (*Zoom completion time*, *Color completion time*, and *Accept completion time*). The only interaction effect we observed was for the Color function - post-hoc analysis with Bonferroni correction showed that the only significant effect was from the handedness for the mid-air interaction technique ($t = -2.64, p = 0.01$). For Zoom technique had a main effect on the completion time. For the Accept function, both technique and handedness had a main effect on the completion time. In both cases, HPUI was better.

Figure 4 shows the summary of these results.

3.7.2 Visual attention

For all attention-related metrics, we conducted the Friedman rank sum test. The effect of the attention-percentage for the factors *technique* and *handedness* can be found in Table 1 under *combined attention-percentage*. A more detailed summary of the results can be found in Figure 5.

Some participants used their peripheral vision to interact with the workspace by placing the menus right beside the workspace and with their gaze fixed on the menus. This resulted in a larger number of trials (7.5%) where they had not looked at the workspace at all. Note that, all participants had re-positioned both the workspace and the mid-air menu position.

Similar to completion time, we also analyzed the attention-percentage for each function individually. Friedman rank sum tests on attention percentage for each function for the factors *technique* and *handedness* show that the technique had a significant effect on all three functions with relatively large effect sizes, except for the Accept function for *handedness*. This can also be seen in Figure 5.

For attention switches, Friedman rank sum tests were conducted for all factors for all functions individually as well as all functions combined (per trial). The results are in Table 1 under attention switches. The only main effect that was observed was for the Accept function, where the technique had a main effect on the number of attention switches. The mean number of switches was below one only for the HPUI DH conditions for the last two blocks (block 6 - 0.95 and block 7 - 0.92).

3.7.3 Errors

We observed three types of errors: (1) By design, the participant would have to interact with the zoom slider only during the first trial in a set. Any interactions with the slider during the remainder of the 3 trials in a block would be due to an error. (2) For each trial, the users have to select the color button only once. When there is more than one selection recognized, it should be interpreted as an error. (3) Errors can be calculated for the accept button as well. Note that this does not differentiate between user errors (e.g., selecting one color when they meant to select a different one) and accidental triggers (e.g., when the user selected one color the adjacent color got selected).

We conducted the Friedman rank sum test with the number of errors per trial with factors technique and handedness. The results can be seen in Table 1 under errors and in Figure 7. There were no significant main effects.

3.7.4 Subjective rating

We first analyzed the inter-item reliability for each category with Cronbach's alpha. Before the analysis, we inverted all reverse-coded items. Appendix B (in supplementary material [2]) lists the individual items and indicates the reverse-coded items. Except for the items for the temporal demand (-1.1), all other categories had alpha above 0.7 (Mental demand - .85, Physical demand - .93, Own Performance - .89, Effort - .88, Frustration - .79). Except for the temporal demand, all agreeing categories were combined by items, resulting in 8 items as seen in Table 1. For each item, we conducted repeated measures ANOVAs with the factors *techniques* and *handedness*. The technique had a significant main effect on all items except Question 9 (the third item of temporal demand). Handedness had a main effect only on the own performance rating. As seen in Table 1, the HPUI is rated to be significantly better than mid-air in all but Question 9 (third question under temporal demand). For handedness, while the DH has a better score on most items, it is significant only for *own performance*.

4 Discussion

We further comment on our findings and discuss the limitations and future work.

4.1 Comparison with mid-air direct interaction

We chose to compare the benefits of HPUI to the common mid-air interaction for head-mounted displays. We had expected that the HPUI would have better performance given the tactile affordance. From the results, we can see that is not the case, HPUI and mid-air have similar performances. This implies a potential trade-off between tactile affordance and visual consistency, particularly, when one is a novice at a given functionality, such as selecting items from a menu. Mid-air interfaces have the benefit of having a visual interface that does not change shape during the interaction, unlike the HPUI which deforms as the thumb reaches targets [40]. We further expand on the difference in outcomes as they relate to various functions in Section 4.2. As seen in Figure 4, with the common functions (i.e., Zoom & Accept), participants were faster with HPUI. A clear advantage of HPUI can be seen in the attention requirement (Section 3.7.2) and the subjective ratings of comfort (Section 3.7.4). From the very initial trials, participants look at HPUI less than half the time (see Figure 5a), and results show a continual increase over time. Particularly with the common functions. This implies

[4]For the interpretation of effect sizes, Cohen's guideline was followed [10]

Table 1: Summary of the results from Study 1 and related statistics. Shows results for the factors Techniques (HPUI & mid-air), and Handedness (DH & NDH), for the last two blocks. Also lists the mean ($\bar{x}$) and standard deviation (sd) for the values of the factors. The better values are highlighted in bold.

Measures	Technique				Handedness			
	statistic	HPUI	mid-air	eff. size	statistic	DH	NDH	eff. size
Completion time	$F(1,15)$	$\bar{x}(sd)$	$\bar{x}(sd)$		$F(1,15)$	$\bar{x}(sd)$	$\bar{x}(sd)$	
Combined	4.13 ○	**10.19s** (1.57s)	10.78s (1.23s)	0.04 ○	9.88 ●	**10.13s** (1.16s)	10.85s (1.59s)	0.06 ○
Zoom	52.01 ●	**1.82s** (0.44s)	2.60s (0.49s)	0.42 ◐	0.69 ○	2.24s (0.67s)	**2.18s** (0.54s)	0.00 ○
Color*	0.13 ○	**1.28s** (0.25s)	1.29s (0.28s)	0.00 ○	4.25 ○	**1.24s** (0.24s)	1.33s (0.29s)	0.02 ○
Accept	5.48 ●	**0.79s** (0.25s)	0.90s (0.23s)	0.05 ○	15.5 ●	**0.78s** (0.20s)	0.92s (0.27s)	0.09 ○
Attention percentage	$\chi^2(1)$	$\bar{x}(sd)$	$\bar{x}(sd)$		$\chi^2(1)$	$\bar{x}(sd)$	$\bar{x}(sd)$	
Combined$_a$	9 ●	**057** (0.18)	0.44 (0.16)	0.56 ◍	4 ●	**0.52** (0.18)	0.49 (0.18)	0.25 ○
Zoom$_a$	9 ●	**0.57** (0.21)	0.38 (0.17)	0.56 ◍	4 ●	**0.46** (0.19)	0.45 (0.21)	0.25 ○
Color$_a$	9 ●	**0.49** (0.17)	0.39 (0.16)	0.56 ◍	4 ●	**0.46** (0.17)	0.42 (0.17)	0.25 ○
Accept$_a$	6.25 ●	**0.67** (0.24)	0.51 (0.22)	0.39 ◐	0.25 ○	**0.60** (0.25)	0.58 (0.24)	0.01 ○
Attention switches	$\chi^2(1)$	$\bar{x}(sd)$	$\bar{x}(sd)$		$\chi^2(1)$	$\bar{x}(sd)$	$\bar{x}(sd)$	
Combined	1 ○	**4.86** (1.72)	5.14 (1.51)	0.06 ○	1 ○	**4.92** (1.35)	5.08 (1.86)	0.06 ○
Zoom	1 ○	**1.16** (0.46)	1.21 (0.46)	0.06 ○	0.25 ○	1.20 (0.41)	**1.17** (0.51)	0.01 ○
Color	0.25 ○	2.61 (0.82)	**2.59** (0.76)	0.01 ○	0 ○	2.61 (0.76)	**2.59** (0.83)	0.01 ○
Accept	6.25 ●	**1.08** (0.79)	1.31 (0.60)	0.39 ◐	0.06 ○	**1.08** (0.52)	1.31 (0.84)	0.00 ○
Errors	$\chi^2(1)$	$\bar{x}(sd)$	$\bar{x}(sd)$		$\chi^2(1)$	$\bar{x}(sd)$	$\bar{x}(sd)$	
per-trial	2.27 ○	0.96 (1.4)	**0.5** (0.87)	0.14 ○	1.33 ○	**0.53** (0.87)	0.93 (1.41)	0.08 ○
Subjective ratings	$F(1,15)$	$\bar{x}(sd)$	$\bar{x}(sd)$		$F(1,15)$	$\bar{x}(sd)$	$\bar{x}(sd)$	
Mental demand	12.22 ●	**3.03** (1.37)	3.88 (1.52)	0.08 ○	1.80 ○	**3.30** (1.60)	3.61 (1.40)	0.01 ○
Physical demand	22.42 ●	**3.14** (1.24)	4.25 (1.31)	0.16 ○	0.88 ○	**3.56** (1.35)	3.83 (1.43)	0.01 ○
Temporal demand								
Q7 - Felt rushed	5.21 ●	**0.70** (0.69)	0.97 (0.73)	0.03 ○	2.31 ○	**0.79** (0.74)	0.88 (0.71)	0.00 ○
Q8 - Preferred pace	8.95 ●	**1.34** (0.17)	1.43 (0.18)	0.05 ○	0.73 ○	**1.37** (0.16)	1.41 (0.20)	0.01 ○
Q9 - Perceived pace	0.12 ○	1.33 (0.30)	**1.30** (0.21)	0.00 ○	2.67 ○	1.35 (0.21)	**1.28** (0.30)	0.01 ○
Own performance $_a$	23.7 ●	**4.82** (1.00)	4.06 (1.11)	0.12 ○	4.63 ●	**4.65** (1.08)	4.23 (1.12)	0.04 ○
Effort	31.81 ●	**3.46** (1.28)	4.45 (1.44)	0.11 ○	0.84 ○	**3.85** (1.46)	4.06 (1.44)	0.00 ○
Frustration	17.75 ●	**2.92** (1.38)	3.81 (1.61)	0.08 ○	1.73 ○	**3.17** (1.59)	3.56 (1.52)	0.01 ○

●: $p < 0.05$, ○: $p > 0.05$, ◍: large effect [4], ◐: moderate effect [4], ○: small effect [4], a: larger is better.

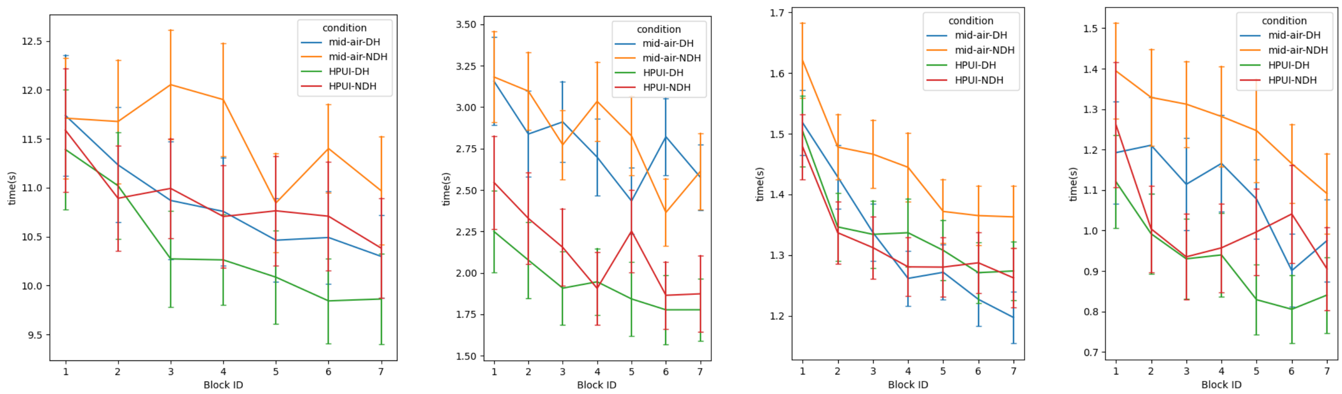
that participants lean on tactile and proprioceptive cues that have been developed over a lifetime - despite this non-static attribute in HPUIs, they have a spatially consistent mapping of commands in relation to thumb-to-finger interactions. This also explains the significant difference in completion time for the Zoom function. As it is a continuous interaction, the tactility of sliding the thumb over the finger allows for completing the interaction more efficiently than interacting with the zoom slider mid-air. Similarly, participants also felt more comfortable while using the HPUI than the mid-air interface. This was also reflected in participant behavior and feedback from the exit questionnaire. When placing the mid-air menu, they would often opt to place it next to the workspace to minimize how much they have to switch gazing back and forth between their primary task and the supporting menu. Participants stated the need to trade comfort for efficiency in doing so.

In summary, our results suggest that the advantage of HPUI appears to be when the participant can rely more on proprioceptive and tactile affordance and not on visual search - and for novice users/functions, where visual search may dominate, there may not be a discernible difference in performance between HPUI and mid-air direct interactions. Hence, HPUI would be preferable for secondary tasks in in-situ scenarios than mid-air direct interactions.

4.2 Analysis of the functions Zoom, Color and Accept

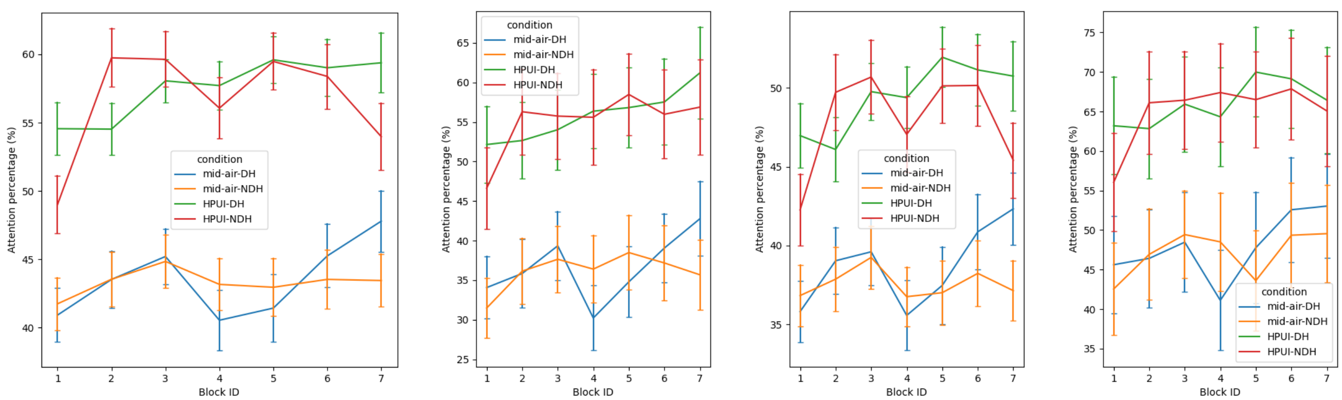
We evaluated the interfaces across three functions: *Zoom*, *Color*, and *Accept*. As stated in Section 3.2, we mapped the Zoom and Accept functions to a specific finger and phalanx. This spatial consistency supports building proprioceptive memory [8]. We observe this effect when considering the performance of the Accept and Color functions. While both functions follow a similar learning trajectory from block 1 to block 7 (see Figure 4 and Figure 6), the Accept function is more efficient and also requires less attention even though they are both the same interaction, i.e., tapping a specific phalanx of a finger. This can also be analyzed through the lens of psychomotor skill acquisition. Following Fitts and Posner's [20] influential three-stage model, the results for the Accept function could be considered to be analogous to the third phase (autonomous phase/expert phase), where the actions are quicker, less attention-demanding, and happen in parallel [8, 44]. However, the results for Color function would be closer to the first phase, (cognitive phase/novice phase), which is slower and relies a lot on deliberate attention and visual search [8, 7]. It is also worth noting that the trajectories of attention percentage (Figure 5) are flatter when compared to completion time and attention switches. We had expected higher attention percentages for the Accept functions. It is unclear if more practice would result in better performance in terms of the attention requirement, where the participants would not have to look at the hand at all for elements like the Accept function. Another potential factor that would have affected this is that participants are switching between novice function, with Color, and expert functions, with Accept. This would be closer to real-world observations - where experts are unaware of or not an expert in all expert functionalities [8]. Hence, momentarily resorting to novice behavior when accessing such functionality, which could have influenced participant behavior.

In summary, the results suggest that the design space of HPUI



(a) Complete set (all functions included) (b) Zoom functions (c) Color function (d) Accept function

Figure 4: Completion time over the blocks separated by the conditions (Technique (HPUI & mid-air), Handedness (DH & NDH)). Shows standard error bars. Figure 4b, Figure 4c and Figure 4d are completion times of the functions zoom, color and accept buttons respectively, averaged on each trial. Figure 4a shows the average time to complete a set.



(a) Complete trial (b) Zoom function (c) Color function (d) Accept function

Figure 5: Attention-percentage (attention on workspace divided by complete time of trial) over the blocks separated by the conditions (Technique (HPUI & mid-air), Handedness (DH & NDH)). Shows standard error bars. Figure 5b, Figure 5c and Figure 5d are attention percentages (while performing a given function attention on workspace divided by the complete time taken for that function in the trial) of functions of zoom, color and accept buttons respectively.

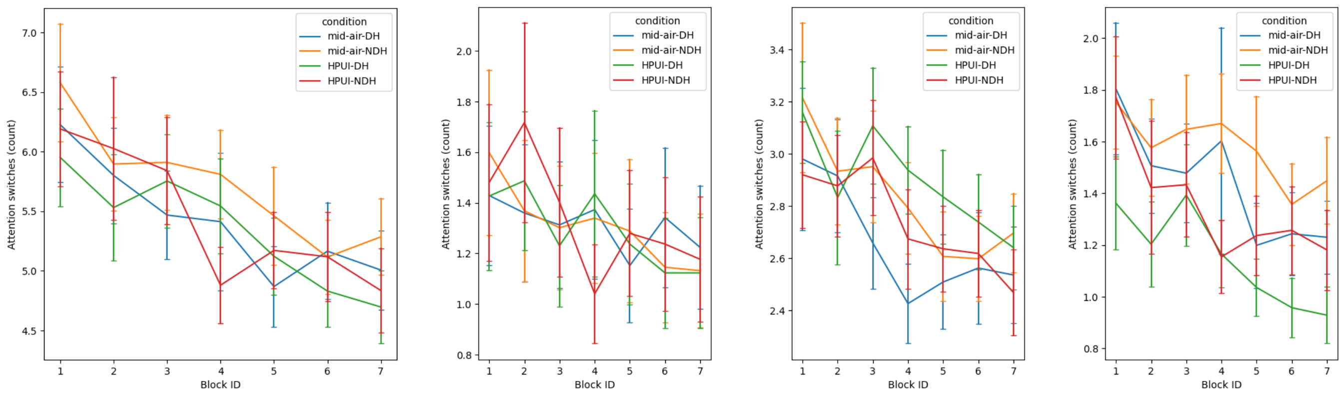
supports novice users with visual cues, while also allowing them to build memory and transition to expert-level level performance. However, one has to be careful with this interpretation, as our study does not directly measure novice and expert behavior. Further research in the context of novice-to-expert transition could help further understand these observations.

Another aspect to consider here is, that we had focused on the on-finger region from design space defined by Faleel et al.,[18]. It is unclear if the off-finger targets would have the same affordance that we have discussed in this paper. The off-finger region in their design space does not have tactile cues. However, with proprioception, one could still make selections in the off-finger space. Future research would have to further evaluate how this compares to on-finger HPUI elements which have tactile cues.

4.3 Use of hand in the secondary interface for dual-tasking scenarios

When considering which hand is preferred to interact with the menu, overall, the results suggest that the DH is preferable for interacting with the supporting task, both in terms of user preference and performance. Since the difference between DH and NDH is not significant across many measures, it is also possible to consider applications that allow the use of either hand based on the situation. For example, the DH could be assigned to the primary tasks, while the NDH to the supporting, digital task. However, in practice, such menus may not be tied to one hand as it may not always be possible to control which hand is available for interaction in an in-situ context. Our results suggest this could be a valid design choice. Further, our results echo prior work on handedness, where the DH is more performant, but can have overlap under some circumstances [13, 2]. Our results also show that the same assumption used by prior work on microgestures being assigned to the NDH [11, 60] would not be applicable when more complex interactions are considered. However, as being forced to use one hand has been showed to have an impact on results [13], one would have to be careful with the above interpretations.

Also, further research is needed to know if the symmetrical nature of hand interactions, such as with the zoom slider in this work, would also translate to other widgets and applications. However, it is worth noting that the errors with the HPUI on the NDH are higher (see Figure 7), which would indicate lower precision with thumb-to-finger interactions on the NDH.



(a) Complete trial (b) Zoom function (c) Color function (d) Accept function

Figure 6: Attention-switches (number of times looking away from workspace and vice versa) over the blocks separated by the conditions (Technique (HPUI & mid-air), Handedness (DH & NDH)). Shows standard error bars. Figure 6b, Figure 6c and Figure 6d are attention switches of functions of zoom, color and accept buttons respectively

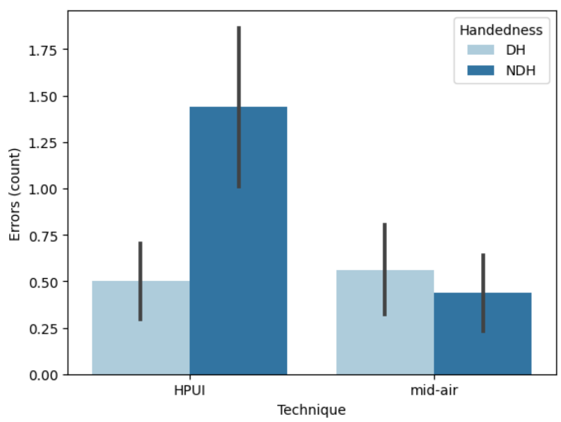


Figure 7: Number of errors made, averaged across trials in the last two blocks and grouped by the technique (HPUI & mid-air) then Handedness (DH & NDH). Shows standard error bars.

With elements such as text, it would be necessary to consider how it would impact performance, comfort, and legibility. Particularly during the novice stages, where they rely on visual search.

4.4 Limitations and Future Work

This paper mainly focuses on analyzing the effect of proprioceptive and tactile affordance of HPUI in in-situ task performance. More precisely, the factors we focus on are efficiency, comfort, and eyes-free affordance. The study focused on a secondary task that is associated with the primary task, which is similar to related work on bi-manual interactions [11, 60]. As opposed to the two independent tasks, we had chosen a compound task design as it allows us to further observe the difference in the functions (i.e., Zoom, Color, and Accept). With independent tasks, the secondary task would have to be more sporadic and/or the primary task more cognitively demanding. This would have affected the results observed in our study. Nonetheless, we expect our conclusions and results related to time and attention would generalize to other types of tasks; as other combinations of tasks also would have a similar outcome - where novice functions/users rely on slower visual search, and lean on proprioception and tactile cues for more common/expert functions which would result in faster performance. But, further research is needed with appropriate in-the-wild studies.

Our results are based on a limited number of 16 participants. In addition, most participants had experience with HMDs and were male. Hence results should be interpreted with care.

Although the factors we explore are based on the literature, more work is needed to validate the influence each of these factors could have on this class of tasks which is well beyond the scope of this paper. Even within the scope of in-situ tasks, there are other combinations one could consider. We had limited our scope of the task to the asynchronous bimanual dependent task to minimize confounding variables. As mentioned above, independent tasks could have different results, but similar conclusions. One could also consider activities that require two hands and have transient widget/menu interactions with one hand. Such interactions would have to also consider factors such as the activation of the transient widgets/menu. These factors need to be fleshed out in future work.

Moreover, when considering non-in-situ tasks, such as working at a desk with a computer, HPUI could be used as another input modality, particularly, for the input of hotkeys. However, the impact HPUI's characteristics and limitations, as discussed by Perella-Holfeld et al. [40], would have on these tasks is unclear and would need further research.

5 Conclusion

We study the advantage of the tactile and proprioceptive affordance of HPUI for in-situ interactions, particularly, involving the use of an interface to support a primary task. Through our evaluation with a compound task where we compared HPUI with mid-air direct interactions, we have drawn the following conclusions: (1) HPUI demonstrates comparable efficiency to mid-air direct interaction and performs significantly better for common interactions, such as selecting a frequent command. (2) Participants reported a higher level of comfort with HPUI. (3) HPUI offers a significant advantage in terms of attention requirement. Once users become familiar with the interface, they tend to pay less attention to the hand and menu, allowing them to focus more on the primary task at hand. (4) Additionally, while both hands can be used effectively for interacting with menus, using the dominant hand for the menus was found to be slightly preferred.

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